

T fixtures – io_trace.sh RED-test corpus (plan.md T-C)

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Contract: specs/004-fast-dev-cycles/plan.md §T-C02 ("Observed-I/O capture and the gate → inputs map") + FR-005/FR-007 in specs/004-fast-dev-cycles/spec.md . Protecting-tests line, verbatim from plan.md: "control needle: a gate known to read CLAUDE.md must list it; golden-bad: a gate that reads through a dynamically built path must still record the resolved path; paired mutation: filter out stat calls → a gate that decides on file existence loses an input and the needle FAILs." tasks.md T052 adds a third, explicitly-named negative control (the CT-5 addition — plan.md line 128): "a gate reading only its own script directory yields exactly that read set and PASSes".

The tool under test (not yet implemented – T-C , a later task)

constitution/scripts/fastcycle/gates/io_trace.sh , per plan.md's path table (line 186: io_trace.sh # observed-I/O capture + gate→inputs map build (T-C02)). Per T-C02's own "Work" line, the real implementation runs each gate under strace -f -e trace=openat,stat,execve in a clean checkout and records every path read and every path written, building a gate → reads/ gate → writes map.

Assumed CLI contract (UNCONFIRMED by the contract itself — no contracts/ file names this tool explicitly; DEFINED here for this RED test's own GREEN-branch, binding-if-adopted on T-C02's implementer, following the house precedent in fixtures/verdict_cache/README.md and fixtures/dispatch_stamp/):

```
io_trace.sh <gate-script> [gate-args...]
```

Runs `<gate-script> [gate-args...]` under strace, and on stdout emits ONE JSON object:

```
{"reads": ["<absolute path>", ...], "writes": ["<absolute path>", ...]}
```

Exit 0 on a successful trace (regardless of the traced gate's own exit code — the tracer's job is to observe, not to judge the gate); nonzero only on a tracer-level failure (gate script missing, strace unavailable, etc).

The scenarios

Directory	Scenario	Protecting-tests line it satisfies
<code>needle_gate.sh</code> (no subdir — single-file fixture)	Control needle	"a gate known to read <code>CLAUDE.md</code> must list it"
<code>dynamic_path_gate/</code>	Golden-bad	"a gate that reads through a dynamically built path must still record the resolved path"
<code>own_dir_only_gate/</code>	Negative control (CT-5)	"a gate reading only its own script directory yields exactly that read set and PASSes"

`needle_gate.sh` – control needle

Reads the REAL project `CLAUDE.md` (never a copy — the needle must be a genuinely known-present value per §11.4.273, and this repo's own top-level `CLAUDE.md` is that value in any checkout of it). `io_trace.sh needle_gate.sh <repo-root>` must list a path ending in `CLAUDE.md` in its `reads` array.

dynamic_path_gate/ – golden-bad

The gate script builds its target file's path from three concatenated shell variables (`base` + `_data` + `ext`) rather than a single literal, so a naive static-analysis (grep-the-source) approach would see only the unexpanded template string, never the real path. `io_trace.sh` MUST record the RESOLVED absolute path ending in `target_data.txt` — this is why T-C02 mandates strace-based OBSERVED tracing (the syscall layer only ever sees post-expansion paths) rather than source scanning.

own_dir_only_gate/ – negative control (CT-)

The gate reads ONLY files inside its own directory: itself (the shell interpreter opens the script file to execute it) and one sibling data file. `io_trace.sh` 's reported read-set for this gate MUST equal EXACTLY `{own_dir_only_gate.sh, sibling_data.txt}` — no extra basenames (the real implementation must filter dynamic-linker/shared-library noise strace also observes) and no omissions. This is the dual false-positive/false-negative guard (§11.4.201(1)) for the tracer mechanism itself, not merely a "does it find something" smoke test.

What this RED test does NOT cover

- The `writes` side of the map (no fixture gate here performs a write) — T-C02's own implementer is expected to add write-side fixtures alongside the reads-side ones landed here, or a follow-up task does.
- The paired mutation named in `plan.md` ("filter out `stat` calls → a gate that decides on file existence loses an input and the needle FAILs") — that is a mutation of the REAL implementation once it exists (T-C02), not something a RED test against an absent tool can exercise; T179's batched mutation sweep is where it belongs.
- Re-tracing on gate-script-hash-change and the "fully on the backstop" cadence rule from T-C02's Work line — those are implementation-internal caching/scheduling behaviour, not I/O-observation-correctness fixtures.